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Connector structure

Abstract

A connector structure (CS) includes an adapter (**1**) to be fitted to an opposing connector (**100**) that includes opposing terminals (**102, 103**) in parallel with each other and a connector (**2**) to be fitted to the adapter (**1**) in a fitting direction (X**1**). The connector includes a connector housing (**3**), a center terminal (**4**), and an outer terminal (**5**). The center terminal (**4**) includes a center terminal portion (**40**) with a central axis (C**1**) extending in the fitting direction. The outer terminal (**5**) includes a cylindrical outer terminal portion (**50**) that concentrically encloses the center terminal portion (**40**). The adapter (**1**) includes an adapter housing (**7**), a first intermediate contact (**8**), and a second intermediate contact (**9**). The adapter housing (**7**) and the connector housing (**3**) are rotatable relative to each other about the central axis (C**1**). The first intermediate contact (**8**) includes a first contact piece portion (**81**) to come into contact with an outer peripheral surface (**40***b*) of a protruding portion (**40***a*) of the center terminal portion (**40**) and a second contact piece portion to come into contact with one of the opposing terminals. The second intermediate contact (**9**) includes a third contact piece portion (**91**) to come into contact with an outer peripheral surface (**50***a*) of the outer terminal portion (**50**) and a fourth contact piece portion (**92**) to come into contact with one of the opposing terminals.

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates to a connector structure.
BACKGROUND ART
(2) A connector disclosed in Patent Literature 1 includes a socket element and a plug element arranged to be connected to each other. The socket element has a pair of pins located side-by-side in parallel with each other, and the pair of pins are each connected to a corresponding wire. Meanwhile, the plug element has a pair of female terminals located side-by-side in parallel with each other, and the pair of female terminals are arranged to be connected, respectively, to the pair of pins.
CITATION LIST
Patent Literature
(3) Patent Literature 1: Japanese Unexamined Patent Publication No. 2008-288122
SUMMARY OF INVENTION
Technical Problem
(4) However, when fitting between the socket element and the plug element, it is necessary to adjust the orientation of the plug element with respect to the socket element so that the pair of pins and the pair of female terminals are aligned with each other. This may result in poor fitting workability.
(5) A preferred embodiment of the present invention provides a connector structure for improved fitting workability with no need to adjust the orientation with respect to an opposing connector.
Solution to Problem
(6) A preferred embodiment of the present invention provides a connector structure including an adapter to be fitted to an opposing connector that includes an opposing housing and a plurality of opposing terminals supported on the opposing housing and extending in parallel with each other, and a connector arranged to be fitted to the adapter in a fitting direction. The connector includes a

connector housing, a center terminal supported on the connector housing and including a pin-shaped center terminal portion with a central axis extending in the fitting direction, and an outer terminal supported on the connector housing and including a cylindrical outer terminal portion that concentrically encloses the center terminal portion with an interposed insulating portion, the center terminal portion includes a protruding portion that protrudes from the outer terminal portion in the fitting direction. The adapter includes an adapter housing to be fitted to the connector housing in the fitting direction and displaceable rotationally relative to the connector housing about the central axis and a first intermediate contact and a second intermediate contact supported on the adapter housing. The first intermediate contact includes a first contact piece portion to elastically contact the outer peripheral surface of the protruding portion of the center terminal portion and a second contact piece portion to elastically contact a corresponding one of the opposing terminals. The second intermediate contact includes a third contact piece portion to elastically contact the outer peripheral surface of the outer terminal portion and a fourth contact piece portion to elastically contact a corresponding one of the opposing terminals.

(7) In accordance with the arrangement above, the adapter is connected to the opposing connector in advance. That is, the second contact piece portion of the first intermediate contact and the fourth contact piece portion of the second intermediate contact of the adapter are connected, respectively, with corresponding ones of the opposing terminals. When fitting the connector into the adapter in the fitting direction, the first contact piece portion of the first intermediate contact elastically contacts the outer peripheral surface of the protruding portion of the center terminal portion of the connector and the third contact piece portion of the second intermediate contact elastically contacts the outer peripheral surface of the outer terminal portion of the connector. Since the adapter housing and the connector housing are displaceable rotationally relative to each other about the central axis of the center terminal portion, the fitting workability is improved with no need to adjust the orientation of the connector with respect to the adapter.

(8) In a preferred embodiment, the connector structure is configured such that, when the adapter housing and the connector housing are displaced relatively and rotationally with respect to each other, a contact portion of the first contact piece portion is displaced circumferentially in a sliding manner with respect to the outer peripheral surface of the center terminal portion and a contact portion of the third contact piece portion is displaced circumferentially in a sliding manner with respect to the outer peripheral surface of the outer terminal portion. In accordance with this arrangement, terminal connections using the adapter and the connector are substantially possible regardless of the orientation with respect to the opposing connector.

(9) In a preferred embodiment, the adapter housing includes a fitting cylinder portion that has an annular engagement protrusion extending entirely circumferentially on the outer peripheral surface of the adapter housing, the connector housing includes a cylindrical portion centered on the central axis, the cylindrical portion split circumferentially with a slit that extends axially to the tip end to be radially and elastically deformable and having at least one locking protrusion on the inner peripheral surface of the cylindrical portion, and the connector structure is configured such that, when the fitting cylinder portion is inserted and fitted into the cylindrical portion, the locking protrusion overrides and engages the engagement protrusion so that the fitting cylinder portion is prevented from coming off.

(10) In accordance with the arrangement above, when fitting between the adapter and the connector, the locking protrusion on the inner peripheral surface of the cylindrical portion on the connector side elastically overrides and engages the annular engagement protrusion on the outer peripheral surface of the fitting cylinder portion on the adapter side, which prevents the adapter and the connector from coming off. When the cylindrical portion on the connector side and the fitting cylinder portion on the adapter side rotate circumferentially relative to each other, the locking protrusion and the engagement protrusion also rotate circumferentially relative to each other. This allows for retention between the adapter and the connector regardless of the orientation with

respect to the opposing connector.
(11) In a preferred embodiment, the plurality of outer terminals are provided in a mutually insulated manner, the plurality of second intermediate contacts are provided correspondingly to the plurality of outer terminals, and a contact portion of each of the plurality of second intermediate contacts and a corresponding one of the outer terminals are positionally offset with respect to each other in the fitting direction. In accordance with this arrangement, fitting regardless of the orientation of the connector is made possible even if there may be three or more poles of opposing terminals and connector terminals.
(12) In a preferred embodiment, the insulating portion includes an insulating member including an insulating cylinder portion into which the center terminal portion is inserted and fitted, and which is inserted and fitted into the outer terminal portion. The insulating cylinder portion is centered on the central axis. In accordance with this arrangement, insulation can be provided between the center terminal portion and the outer terminal portion, which are located concentrically, with such a simple structure using the insulating cylinder portion.
(13) In a preferred embodiment, the center terminal includes a first wire connecting portion extending from the center terminal portion and to which a corresponding wire is connected, the outer terminal includes a second wire connecting portion extending from the outer terminal portion and to which a corresponding wire is connected, and the insulating member includes an insulating wall portion extending from the insulating cylinder portion and providing insulation between the first wire connecting portion and the second wire connecting portion. In accordance with this arrangement, insulation can be provided between the first wire connecting portion and the second wire connecting portion with such a simple structure using the insulating wall portion that is provided in a manner extending from the insulating cylinder portion.
(14) In a preferred embodiment, the center terminal, the outer terminal, and the insulating member may be unitized. In accordance with this arrangement, assembling work can be facilitated.
(15) In a preferred embodiment, the adapter housing includes a to-be-positioned portion to engage a positioning portion of the opposing housing so that the first intermediate contact and the second intermediate contact are aligned, respectively, with corresponding ones of the opposing terminals. In accordance with this arrangement, when preliminarily fitting the adapter with respect to the opposing connector, the intermediate contacts are easily positioned with respect to corresponding ones of the opposing terminals.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1**A is a schematic perspective view of a state before fitting between a connector and an adapter fitted preliminarily to an opposing connector of a connector structure according to a preferred embodiment of the present invention.
(2) FIG. **1**B is a schematic perspective view of the connector structure in a state fitted to the opposing connector.
(3) FIG. **2** is a cross-sectional view of the connector, as a component of the connector structure.
(4) FIG. **3** is a cross-sectional view of the connector structure in a fitting state.
(5) FIG. **4** is an exploded perspective view of the connector.
(6) FIG. **5**A is a perspective view of a center terminal of the connector, FIG. **5**B is a plan view of the center terminal, and FIG. **5**C is a side view of the center terminal.
(7) FIG. **6**A is a perspective view of an outer terminal of the connector, FIG. **6**B is a plan view of the outer terminal, and FIG. **6**C is a side view of the outer terminal.
(8) FIG. **7**A is a perspective view of an insulating member, FIG. **7**B is a front view of the insulating member, and FIG. **7**C is a rear view of the insulating member.

(9) FIG. **8**A is a schematic perspective view of the connector, FIG. **8**B is a front view of the connector, and FIG. **8**C is a side view of the connector.
(10) FIGS. **9**A, **9**B, and **9**C are a perspective view, a front view, and a rear view, respectively, of a connector housing in an open state.
(11) FIG. **10** is an exploded perspective view of the adapter.
(12) FIG. **11** is a perspective view of the adapter.
(13) FIG. **12**A is a perspective view of a first intermediate contact, as a component of the adapter, and FIGS. **12**B and **12**C are side views of the first intermediate contact when viewed in their respective different directions.
(14) FIGS. **13**A and **13**B are a perspective view and a side view, respectively, of a second intermediate contact, as a component of the adapter.
(15) FIGS. **14**A, **14**B, and **14**C are a front view, a rear view, and a side view, respectively, of the adapter.
(16) FIG. **15** is an enlarged front view of the adapter in a state attached to the opposing connector.
(17) FIG. **16**A is a cross-sectional view of the adapter in a state attached to the opposing connector, taken along line **16**A-**16**A in FIG. **15**.
(18) FIG. **16**B is a cross-sectional view of the adapter in a state attached to the opposing connector, taken along line **16**B-**16**B in FIG. **15**.
(19) FIGS. **17**A, **17**B, and **17**C are perspective views of the connector structure in a state fitted to the opposing connector in their respective different orientations.
(20) FIG. **18** is a schematic cross-sectional view of a center terminal and an outer terminal according to a modification.
DESCRIPTION OF EMBODIMENTS
(21) Preferred embodiments embodying the present invention will hereinafter be described with reference to the accompanying drawings.
(22) FIGS. **1**A and **1**B are schematic perspective views showing a connector structure in use according to a preferred embodiment of the present invention. The connector structure CS includes an adapter **1** fitted preliminarily to an opposing connector **100** and a connector **2** arranged to be fitted to the adapter **1** in a fitting direction X**1**.
(23) FIG. **2** is a cross-sectional view of the connector **2** and FIG. **3** is a cross-sectional view of the connector structure CS in a fitting state. As shown in FIG. **3**, the opposing connector **100** includes an opposing housing **101** and a plurality of (for example, two) opposing terminals **102** and **103**. The two opposing terminals **102**, **103** are pin-shaped terminals housed and retained in the opposing housing **101** and extending in parallel with each other in the fitting direction X**1**.
(24) FIG. **4** is an exploded perspective view of the connector **2**. As shown in FIG. **4**, the connector **2** includes a connector housing **3**, a center terminal **4**, an outer terminal **5**, and an insulating member **6**. The connector housing **3** is formed of an insulating material. The center terminal **4** and the outer terminal **5** are formed of a conductive material. The center terminal **4** and the outer terminal **5** are supported on the connector housing **3**. In FIG. **4**, a main body portion **30** and a cover **32**, which are components of the connector housing **3**, are shown in a separated manner.
(25) As shown in FIGS. **2** to **4**, the center terminal **4** includes a pin-shaped center terminal portion **40** extending in a fitting direction X**1**. The center terminal portion **40** may have a hollow pin shape as shown. The center terminal portion **40** has a central axis C**1**. The outer terminal **5** includes a cylindrical outer terminal portion **50** that concentrically encloses the center terminal portion **40** with the interposed insulating member **6**. That is, the outer terminal portion **50** shares the central axis C**1** with the center terminal portion **40**. The center terminal portion **40** includes a protruding portion **40** that protrudes from the outer terminal portion **50** in the fitting direction X**1**.
(26) Specifically, the insulating member **6** is engaged with the connector housing **3** in an engageable and disengageable manner. The insulating member **6** is interposed between the center terminal **4** and the outer terminal **5** to provide insulation between the center terminal **4** and the

outer terminal **5**.
(27) The adapter **1** includes an adapter housing **7**, a first intermediate contact **8**, and a second intermediate contact **9**. The adapter housing **7** is formed of an insulating material. The adapter housing **7** is arranged to be fitted to the connector housing **3** in the fitting direction X**1**. The adapter housing **7** is displaceable relatively and rotationally with respect to the connector housing **3** about the central axis C**1** of the center terminal **4**.
(28) The first intermediate contact **8** and the second intermediate contact **9** are formed of a conductive material and supported on the adapter housing **7**.
(29) The first intermediate contact **8** includes a fixed portion **80**, a first contact piece portion **81**, and a second contact piece portion **82**. The fixed portion **80** is pressed and fixed into the adapter housing **7**. The first contact piece portion **81** is provided in a manner cantilevered and extending on/from the fixed portion **80** and arranged to elastically contact the outer peripheral surface **40***b* of the protruding portion **40***a* of the center terminal portion **40**. The second contact piece portion **82** is provided in a manner cantilevered and extending on/from the fixed portion **80** and arranged to elastically contact the corresponding opposing terminal **102**.
(30) The second intermediate contact **9** includes a fixed portion **90**, a third contact piece portion **91**, and a fourth contact piece portion **92**. The fixed portion **90** is pressed and fixed into the adapter housing **7**. The third contact piece portion **91** is provided in a manner cantilevered and extending on/from the fixed portion **90** and arranged to elastically contact the outer peripheral surface **50***a* of the outer terminal portion **50**. The fourth contact piece portion **92** is provided in a manner cantilevered and extending on/from the fixed portion **90** and arranged to elastically contact the corresponding opposing terminal **103**.
(31) Next, the center terminal **4** will be described.
(32) FIG. **5**A is a perspective view of the center terminal **4**, FIG. **5**B is a plan view of the center terminal **4**, and FIG. **5**C is a side view of the center terminal **4**.
(33) As shown in FIGS. **5**A to **5**C, the center terminal **4** includes a center terminal portion **40**, a support portion **41**, a tapered portion **42**, a positioning wall portion **43**, and a first wire connecting portion **44**. The center terminal **4** is formed by using sheet metal. The center terminal portion **40** and the support portion **41** are formed in a cylindrical shape about the central axis C**1**. The center terminal portion **40** is provided in a manner extending from an axial end of the support portion **41** through the tapered portion **42** to have a diameter smaller than that of the support portion **41**.
(34) A rear half portion of the support portion **41** (a portion opposite to the center terminal portion **40** side) is cut open. The first wire connecting portion **44** is provided in a manner extending downward from the cut-open portion of the support portion **41** through the positioning wall portion **43** orthogonally to the central axis C**1**. Specifically, the first wire connecting portion **44** includes a wire barrel **44***a* to which the core of a wire D**1** (see FIG. **4**) is connected and an insulation barrel **44***b* to which the insulating coating portion of the wire D**1** is connected.
(35) The positioning wall portion **43** includes a pair of wall portions **43***a*, **43***b* opposing each other in a direction parallel with the central axis C**1** and a pair of wall portions **43***c*, **43***d* opposing each other in a direction orthogonal to the central axis C**1**.
(36) Next, the outer terminal **5** will be described.
(37) FIG. **6**A is a perspective view of the outer terminal **5**, FIG. **6**B is a plan view of the outer terminal **5**, and FIG. **6**C is a side view of the outer terminal **5**.
(38) As shown in FIGS. **6**A to **6**C, the outer terminal **5** includes an outer terminal portion **50**, a positioning wall portion **51**, a second wire connecting portion **52**, and a positioning protrusion **53**. The outer terminal **5** is formed by using sheet metal. The outer terminal portion **50** has a cylindrical shape centered on the central axis C**1**. The positioning wall portion **51** is formed by cut-opening an axial end of the outer terminal portion **50**. The positioning wall portion **51** includes a pair of wall portions **51***a*, **51***b* opposing each other in a direction parallel with the central axis C**1** and a pair of wall portions **51***c*, **51***d* opposing each other in a direction orthogonal to the central axis C**1**.

(39) The second wire connecting portion **52** is provided in a manner extending downward from the outer terminal portion **50** through the positioning wall portion **51** orthogonally to the central axis C**1**. Specifically, the second wire connecting portion **52** includes a wire barrel **52***a* to which the core of a wire D**2** (see FIG. **4**) is connected and an insulation barrel **52***b* to which the insulating coating portion of the wire D**2** is connected.
(40) Next, the insulating member **6** will be described.
(41) FIG. **7**A is a perspective view of the insulating member **6**, FIG. **7**B is a front view of the insulating member **6**, and FIG. **7**C is a rear view of the insulating member **6**.
(42) As shown in FIGS. **7**A to **7**C, the insulating member **6** includes an insulating cylinder portion **60** and an insulating wall portion **61**. The insulating cylinder portion **60** mainly functions to provide insulation between the center terminal portion **40** of the center terminal **4** and the outer terminal portion **50** of the outer terminal **5**. The insulating wall portion **61** mainly functions to provide insulation between the first wire connecting portion **44** of the center terminal portion **40** of the center terminal **4** and the second wire connecting portion **52** of the outer terminal **5**.
(43) Specifically, the insulating cylinder portion **60** includes a peripheral wall portion **60***a*, an end wall portion **60***b*, an outer peripheral surface **60***c*, an inner peripheral surface **60***d*, and an insertion hole **60***e*. The end wall portion **60***b* is located at one end of the insulating cylinder portion **60**. The insertion hole **60***e* is formed in the end wall portion **60***b*.
(44) As shown in FIG. **2**, the center terminal portion **40** and the support portion **41** of the center terminal **4** are inserted and fitted into the insulating cylinder portion **60**. The insulating cylinder portion **60** is inserted and fitted into the outer terminal portion **50** of the outer terminal **5**. One end of the outer terminal portion **50** is aligned with one end of the insulating cylinder portion **60**.
(45) The inner peripheral surface **60***d* of the insulating cylinder portion **60** is shaped to include a small diameter portion along the outer peripheral surface **40***b* of the center terminal portion **40**, a large diameter portion along the outer peripheral surface of the support portion **41**, and a tapered portion adjacent to the tapered portion **42**. A part of the center terminal portion **40** protrudes externally from the insulating cylinder portion **60** (i.e. from the outer terminal portion **50**) through the insertion hole **60***e* to form the protruding portion **40***a*.
(46) As shown in FIGS. **7**A to **7**C, the insulating wall portion **61** includes a top wall portion **62**, a front wall portion **63**, a pair of side wall portions **64**, **65**, a pair of engagement protrusions **66**, and a housing portion **67**. The top wall portion **62** has a semi-cylindrical shape and connects the upper ends of the pair of side wall portions **64**, **65** in an inverted U shape. The housing portion **67** is formed by the top wall portion **62**, the front wall portion **63**, and the pair of side wall portions **64**, **65**, and mainly houses the positioning wall portion **43** and the first wire connecting portion **44** of the center terminal **4** (see FIG. **2**). The front wall portion **63** is interposed between the first wire connecting portion **44** of the center terminal **4** and the second wire connecting portion **52** of the outer terminal **5** and functions to provide insulation between the wire connecting portions **44**, **52**.
(47) The insulating cylinder portion **60** is provided in a manner extending frontward from the front wall portion **63**. The interior of the insulating cylinder portion **60** is in communication with the interior of the housing portion **67** through an insertion hole that runs through the front wall portion **63**.
(48) As shown in FIGS. **7**C, a convex portion **64***b* is formed in a manner protruding from the inner surface **64***a* of the side wall portion **64**. Within the housing portion **67**, the inner surfaces of the top wall portion **62**, the front wall portion **63**, and the pair of side wall portions **64**, **65** and the upper surface of the convex portion **64***b* define a retaining portion **68** arranged to fit and retain the positioning wall portion **43** of the center terminal **4**.
(49) Specifically, in a state where the center terminal **4** shown in FIGS. **5**A to **5**C is attached to the insulating member **6** shown in FIGS. **7**A to **7**C, the center terminal portion **40** and the support portion **41** of the center terminal **4** are inserted through the insulating cylinder portion **60** so that a part of the center terminal portion **40** protrudes from the insulating cylinder portion **60** as the

protruding portion **40**. The wall portion **43***c* of the positioning wall portion **43** of the center terminal **4** is laid along the inner surface **64***a* of the side wall portion **64** of the insulating wall portion **61** of the insulating member **6**, and the lower edge of the wall portion **43***c* is received by the upper surface **64***c* of the convex portion **64***b*. The wall portion **43***d* of the positioning wall portion **43** of the center terminal **4** is laid along the inner surface **65***a* of the side wall portion **65** of the insulating wall member **6**. The wall portion **43***a* of the positioning wall portion **43** of the center terminal **4** is received by the inner surface **63***a* (corresponding to the back surface) of the front wall portion **63** of the insulating wall portion **61** of the insulating member **6**. The center terminal **4** is thus positioned with respect to the insulating member **6** in three orthogonal directions including the direction of the central axis C**1**.
(50) A recessed portion **63***c* is formed in a manner protruding from the outer surface **63***b* (corresponding to the front surface) of the front wall portion **63**. In a state where the outer terminal **5** shown in FIGS. **6**A to **6**C is attached to the insulating member **6** shown in FIGS. **7**A to **7**C, the outer terminal portion **50** of the outer terminal **5** is fitted externally to the insulating cylinder portion **60** of the insulating member **6**. Also, the wall portion **51***b* of the positioning wall portion **51** of the outer terminal **5** is received by the outer surface **63***b* of the front wall portion **63** of the insulating wall portion **61** of the insulating member **6**.
(51) The positioning protrusion **53** of the outer terminal **5** is inserted and fitted into a corresponding engagement groove **30***h* of the connector housing **3** (see FIG. **9***c*).
(52) Next, the connector housing **3** will be described.
(53) FIGS. **8**A, **8**B, and **8**C are a schematic perspective view, a front view, and a side view, respectively, of the connector **2**. FIGS. **9**A, **9**B, and **9**C are a perspective view, a front view, and a rear view, respectively, of the connector housing **3** in an open state.
(54) Referring to FIGS. **8**A, **8**B, **8**C, **9**A, **9**B, and **9**C, the connector housing **3** includes a main body portion **30**, a cylindrical portion **31**, a cover **32**, and a flexible connecting portion **33**. The main body portion **30** has an approximately rectangular parallelepiped shape and includes a top wall portion **30***a*, a pair of side wall portions **30***b*, and a front wall portion **30***c*. The main body portion **30** is opened rearward and the rear opening portion of the main body portion **30** is closed by the cover **32** in an assembled state.
(55) The cylindrical portion **31** is centered on a central axis that coincides with the central axis C**1**. The cylindrical portion **31** is split circumferentially with a plurality of slits **31***a* that extend axially to the tip end to be radially and elastically deformable. A locking protrusion **31***c* is formed in a manner protruding from the inner peripheral surface **31***b* of the cylindrical portion **31** (see FIGS. **8**A, **8**B).
(56) At least one locking protrusion **31***c* may be provided. In the example of FIG. **8**B, four locking protrusions **31***c* are provided at circumferentially regular intervals. The locking protrusions **31***c* are arranged to elastically override and engage an annular engagement protrusion **74** (see FIG. **10**) of the adapter **1**.
(57) The flexible connecting portion **33** connects the rear edge of the upper wall portion **30***a* of the main body portion **30** and the cover **32** to serve as a hinge when the cover **32** is rotationally opened and closed. The cover **32** includes a plate-shaped main body portion **32***a* and two pairs of left and right U-shaped elastically deformable grappling arms **32***b*, **32***c*.
(58) Two pairs of engagement protrusions **30***d*, **30***e* arranged to be engaged, respectively, by the corresponding pairs of grappling arms **32***b*, **32***c* of the cover **32** are formed in a manner protruding from the pair of side wall portions **30***b* of the main body portion **30**. The grappling arms **32***b*, **32***c* are arranged to elastically override and engage, respectively, the corresponding engagement protrusions **30***d*, **30***e* when the cover **32** is closed. This causes the cover **32** to be locked in a closed state.
(59) An elastic hook portion **30***f* (see FIG. **4**) arranged to engage a corresponding one of the engagement protrusions **66** provided on the insulating member **6** is provided at the rear edge of

each side wall portion **30***b* of the main body portion **30**. When each elastic hook portion **30***f* engages the corresponding engagement protrusion **66** of the insulating member **6**, the insulating member **6** is retained within the connector housing **3**.
(60) As shown in FIGS. **9**C, a terminal portion housing portion **34** and a connecting space portion **35** are defined within the main body portion **30**. The terminal portion housing portion **34** mainly houses the center terminal portion **40** and the outer terminal portion **50** (see FIG. **2**). The connecting space portion **35** mainly houses the first wire connecting portion **44** of the center terminal **4** and the second wire connecting portion **52** of the outer terminal **5**, and the wire connecting portions **44**, **52** are connected, respectively, with the corresponding wires D**1**, D**2** within the connecting space portion **35**.
(61) The terminal portion housing portion **34** is defined by an inverted U-shaped first delimiting wall portion **30***g* provided in the main body portion **30**. An engagement groove **30***h* with which the positioning protrusion **53** of the outer terminal **5** is engaged is formed in a top portion of the first delimiting wall portion **30***g*. An insertion hole **30***j* through which the terminal portion housing portion **34** is opened externally is formed in the front wall portion **30***c* of the main body portion **30**. The center terminal portion **40**, the insulating cylinder portion **60**, and the outer terminal portion **50** partially protrudes externally through the insertion hole **30***j*.
(62) The connecting space portion **35** is located below the terminal portion housing portion **34** in communication with each other. The wires D**1**, D**2** drawn from below into the connecting space portion **35** (see FIG. **2**) are connected, respectively, to the first wire connecting portion **44** of the center terminal **4** and the second wire connecting portion **52** of the outer terminal **5** within the connecting space portion **35** corresponding thereto. Also, within the connecting space portion **35**, a convex portion **30***m* is formed in a manner protruding from the front wall portion **30***c*. The wall portion **51***c* of the positioning wall portion **51** of the outer terminal **5** inserted within the terminal portion housing portion **34** is received by the upper surface **30***n* of the lower convex portion **30***m*. A punched hole **30***k* is formed in the convex portion **30***m*.
(63) Next, the adapter **1** will be described.
(64) FIG. **10** is an exploded perspective view of the adapter **1**. FIG. **11** is a perspective view of the adapter **1**. FIG. **12**A is a perspective view of the first intermediate contact **8** and FIGS. **12**B and **12**C are side views of the first intermediate contact **8** when viewed in their respective different directions. FIGS. **13**A and **13**B are a perspective view and a side view, respectively, of the second intermediate contact **9**. FIGS. **14**A, **14**B, and **14**C are a front view, a rear view, and a side view, respectively, of the adapter **1**. FIG. **15** is an enlarged front view of the adapter **1** in a state attached to the opposing connector **100**. FIG. **16**A is a cross-sectional view of the adapter **1** in a state attached to the opposing connector, taken along line **16**A-**16**A in FIG. **15**. FIG. **16**B is a cross-sectional view of the adapter **1** in a state attached to the opposing connector **100**, taken along line **16**B-**16**B in FIG. **15**.
(65) As shown in FIG. **10**, the adapter **1** includes an adapter housing **7**, a first intermediate contact **8**, and a second intermediate contact **9**. The first intermediate contact **8** is arranged to be connected with the opposing terminal **102** and the second intermediate contact **9** is arranged to be connected with the opposing terminal **103**.
(66) The first intermediate contact **8** will first be described.
(67) As shown in FIGS. **10**, **12**A, **12**B, and **12**C, the first intermediate contact **8** includes a fixed portion **80**, a first contact piece portion **81**, and a second contact piece portion **82**. The fixed portion **80** includes a first fixed piece portion **80***a*, a second fixed piece portion **80***b*, and a bridge portion **80***c*. The bridge portion **80***c* provides connection between one ends of the first fixed piece portion **80***a* and the second fixed piece portion **80***b*. The first fixed piece portion **80***a* is formed wider than the second fixed piece portion **80***b*. Press-fit protrusions **80***d* are formed in a manner protruding from the outside surfaces of the first fixed piece portion **80***a* and the second fixed piece portion **80***b*.

(68) The first contact piece portion **81** is provided in a manner cantilevered and extending on/from one end of the first fixed piece portion **80***a*. The first contact piece portion **81** includes a first slope portion **81***a*, a second slope portion **81***b*, and a contact portion **81***c*. The first slope portion **81***a* and the second slope portion **81***b* are sloped in opposite directions to form an angle, and the contact portion **81***c* is provided at the top portion of the angle. The contact portion **81***c* of the first contact piece portion **81** is arranged to elastically contact the outer peripheral surface **40***b* of the protruding portion **40***a* of the center terminal portion **40** of the connector **2**.
(69) The second contact piece portion **82** is provided in a manner cantilevered and extending on/from the bridge portion **80***c* to the opposite side of the first contact piece portion **81**. The second contact piece portion **82** includes a first slope portion **82***a*, a second slope portion **82***b*, a contact portion **82***c*, and an extending piece portion **82***d*. The first slope portion **82***a* and the second slope portion **82***b* are sloped in opposite directions to form an angle, and the contact portion **82***c* is provided at the top portion of the angle. The contact portion **82***c* of the second contact piece portion **82** protrudes on the same side as the contact portion **81***c* of the first contact piece portion **81**. The extending piece portion **82***d* is provided in a manner extending from the tip end of the second slope portion **82***b* approximately orthogonally to the fixed portion **80**.
(70) Next, the second intermediate contact **9** will be described.
(71) As shown in FIGS. **10**, **13**A, and **13**B, the second intermediate contact **9** includes a fixed portion **90**, a third contact piece portion **91**, and a fourth contact piece portion **92**. The fixed portion **90** includes a first fixed piece portion **90***a*, a second fixed piece portion **90***b*, and a bridge portion **90***c*. The bridge portion **90***c* provides connection between one ends of the first fixed piece portion **90***a* and the second fixed piece portion **90***b*. The first fixed piece portion **90***a* is formed wider than the second fixed piece portion **90***b*. Press-fit protrusions **90***d* are formed in a manner protruding from the outside surfaces of the first fixed piece portion **90***a* and the second fixed piece portion **90***b*.
(72) The third contact piece portion **91** is provided in a manner cantilevered and extending on/from one end of the first fixed piece portion **90***a*. The third contact piece portion **91** includes a first slope portion **91***a*, a second slope portion **91***b*, and a contact portion **91***c*. The first slope portion **91***a* and the second slope portion **91***b* are sloped in opposite directions to form an angle, and the contact portion **91***c* is provided at the top portion of the angle. The contact portion **91***c* of the third contact piece portion **91** is arranged to elastically contact the outer peripheral surface **50***a* of the outer terminal portion **50** of the connector **2**.
(73) The fourth contact piece portion **92** is provided in a manner cantilevered and extending on/from the bridge portion **90***c* to the opposite side of the third contact piece portion **91**. The fourth contact piece portion **92** includes a first slope portion **92***a*, a second slope portion **92***b*, a contact portion **92***c*, and an extending piece portion **92***d*. The first slope portion **92***a* and the second slope portion **92***b* are sloped in opposite directions to form an angle, and the contact portion **92***c* is provided at the top portion of the angle. The contact portion **92***c* of the fourth contact piece portion **92** protrudes on the same side as the contact portion **91***c* of the third contact piece portion **91**. The extending piece portion **92***d* is provided in a manner extending from the tip end of the second slope portion **92***b* approximately orthogonally to the fixed portion **90**.
(74) Next, the adapter housing **7** will be described with reference to FIGS. **10**, **11**, **14**A, **14**B, **14**C, **15**, **16**A, and **16**B. The adapter housing **7** includes a first end portion **7***a*, a second end portion **7***b*, an outer peripheral portion **7***c*, a first terminal insertion recessed portion **71**, a pair of second terminal insertion recessed portions **72**, a fitting cylinder portion **73**, an annular engagement protrusion **74**, a first intermediate contact retaining groove **75**, a second intermediate contact retaining groove **76**, an annular convex portion **77**, a pair of elastic piece portions **78**, and a pair of to-be-positioned portions **69**.
(75) The outer peripheral portion **7***c* of the adapter housing **7** is formed in an approximately cylindrical shape. A predetermined area from the first end portion **7***a* of the adapter housing **7** in the

fitting direction X**1** forms a cylindrical fitting cylinder portion **73** arranged to be inserted and fitted into the cylindrical portion **31** of the connector **2** side. The first terminal insertion recessed portion **71** is an inside space of the fitting cylinder portion **73** and opened through the first end portion **7***a* side. The center terminal portion **40** and the outer terminal portion **50** of the connector **2** are arranged to be inserted into the first terminal insertion recessed portion **71**.
(76) A circumferentially extending annular engagement protrusion **74** is formed in a manner protruding from the outer peripheral surface of the fitting cylinder portion **83** in the first end portion **7***a*. The locking protrusion **31***c* (see FIGS. **8**B, **9**A, and **9**B) of the connector **2** side is arranged to elastically override and engage the annular engagement protrusion **74** when fitting between the adapter **1** and the connector **2** (see FIG. **3**).
(77) The pair of second terminal insertion recessed portions **72**, into which the pair of respective opposing terminals **102**, **103** are arranged to be inserted, are formed within a predetermined area from the second end portion **7***b* in the direction opposite to the fitting direction X**1** (see FIGS. **16**A and **16**B). The pair of second terminal insertion recessed portions **72** are opened through the second end portion **7***b* side.
(78) The first intermediate contact retaining groove **75** and the second intermediate contact retaining groove **76** are formed within a predetermined area from the second end portion **7***b* side in the direction opposite to the fitting direction X**1** and opened through the second end portion **7***b* side.
(79) The first intermediate contact retaining groove **75** is located adjacent to the first terminal insertion recessed portion **71** into which the center terminal portion **40** is arranged to be inserted and to the second terminal insertion recessed portion **72** into which the one opposing terminal **102** is arranged to be inserted. The fixed portion **80** of the first intermediate contact **8** is pressed and fixed into the first intermediate contact retaining groove **75** (see FIG. **16**A). The contact portion **81***c* of the first contact piece portion **81** of the first intermediate contact **8** is located to advance into the first terminal insertion recessed portion **71**. The contact portion **82***c* of the second contact piece portion **82** of the first intermediate contact **8** is located to advance into the corresponding second terminal insertion recessed portion **72**.
(80) The second intermediate contact retaining groove **76** is located adjacent to the first terminal insertion recessed portion **71** into which the outer terminal portion **50** is arranged to be inserted and to the second terminal insertion recessed portion **72** into which the other opposing terminal **102** is arranged to be inserted. The fixed portion **90** of the second intermediate contact **9** is pressed and fixed into the second intermediate contact retaining groove **76** (see FIG. **16**B). The contact portion **91***c* of the third contact piece portion **91** of the second intermediate contact **9** is located to advance into the first terminal insertion recessed portion **71**. The contact portion **92***c* of the fourth contact piece portion **92** of the second intermediate contact **9** is located to advance into the corresponding second terminal insertion recessed portion **72**.
(81) As shown in FIGS. **10**, **11**, and **14**C, a circumferentially extending annular convex portion **77** is formed in a manner protruding from the outer peripheral portion **7***c* within a predetermined area from the second end portion **7***b*. A pair of cantilevered elastic piece portions **78** are formed in a manner extending by a predetermined length toward the first end portion **7***a* side of the annular convex portion **77** from an end portion closer to the first end portion **7***a* side. The pair of elastic piece portions **78** have an arc-shaped cross section and are located on radially opposite sides. A clearance gap S (see FIG. **16**A) is provided between the pair of elastic piece portions **78** and the outer peripheral portion **7***c*, so that the pair of elastic piece portions **78** are radially and elastically deformable.
(82) A pair of to-be-positioned portions **79** are formed in a manner protruding from the outer surface in the vicinity of a tip end portion of the pair of elastic piece portions **78**. The pair of to-be-positioned portions **79** are each formed in an angled convex portion. The pair of to-be-positioned portions **79** are located in mutually asymmetric positions with respect to the center C**2** of the first

terminal insertion recessed portion **71** (see FIG. **14**A). As shown in FIG. **15**, when the pair of to-be-positioned portions **79** engage the positioning grooves **104**, **105** (positioning portions) provided in the opposing housing **101** of the opposing connector **100**, the adapter **1** is positioned in the circumferential direction of the fitting cylinder portion **73** with respect to the opposing connector **100**.

(83) In accordance with this embodiment, the adapter **1** is connected to the opposing connector **100** in advance, as shown in FIG. **1**A. That is, the second contact piece portion **82** of the first intermediate contact **8** and the fourth contact piece portion **92** of the second intermediate contact **9** of the adapter **1** are connected, respectively, with the corresponding opposing terminals **102**, **103** (see FIGS. **16**A and **16**B). When fitting the connector **2** into the adapter **1** in the fitting direction X**1**, the first contact piece portion **81** of the first intermediate contact **8** elastically contacts the outer peripheral surface **40***b* of the protruding portion **40***a* of the center terminal portion **40** of the connector **2** and the third contact piece portion **91** of the second intermediate contact **9** elastically contacts the outer peripheral surface **50***a* of the outer terminal portion **50** of the connector **2**, as shown in FIG. **3**.

(84) Since the adapter housing **7** and the connector housing **3** are displaceable rotationally relative to each other about the central axis C**1** of the center terminal portion **40**, the connector **2** can be fitted to the adapter **1** no matter how the connector **2** is oriented with respect to the adapter **1**, as shown in FIGS. **1**B and **17**A to **17**C. That is, the fitting workability is improved with no need to adjust the orientation of the connector **2** with respect to the adapter **1**.

(85) Also, referring to FIG. **3**, when the adapter housing **7** and the connector housing **3** are displaced relatively and rotationally about the central axis C**1**, the contact portion **81***c* of the first contact piece portion **81** of the first intermediate contact **8** is displaced circumferentially in a sliding manner with respect to the outer peripheral surface **40***b* of the protruding portion **40***a* of the center terminal portion **40** and the contact portion **91***c* of the third contact piece portion **91** of the second intermediate contact **9** is displaced circumferentially in a sliding manner with respect to the outer peripheral surface **50***a* of the outer terminal portion **50**. This makes terminal connections using the adapter **1** and the connector **2** substantially possible regardless of the orientation with respect to the opposing connector **100**.

(86) Also, when fitting between the adapter **1** and the connector **2**, the locking protrusion **31***c* on the inner peripheral surface **31***b* of the cylindrical portion **31** of the connector **2** side elastically overrides and engages the annular engagement protrusion **74** on the outer peripheral surface of the fitting cylinder portion **73** of the adapter **1** side, as shown in FIG. **3**, which prevents the connector **2** from coming off from the adapter **1**. When the cylindrical portion **31** of the connector **2** side and the fitting cylinder portion **73** of the adapter **1** side rotate circumferentially relative to each other, the locking protrusion **31***c* and the engagement protrusion **74** also rotate circumferentially relative to each other. This allows for retention between the adapter **1** and the connector **2** regardless of the orientation with respect to the opposing connector **100**. In other words, the locking structure cannot interfere with the relative rotational displacement between the adapter **1** and connector **2**.

(87) Also, as shown in FIG. **2**, the insulating member **6** is centered on the central axis C**1** and includes the insulating cylinder portion **60** into which the center terminal portion **40** is inserted and fitted, and which is inserted and fitted into the outer terminal portion **50**. As a result, insulation can be provided between the center terminal portion **40** and the outer terminal portion **50**, which are located concentrically, with such a simple structure using the insulating cylinder portion **60**.

(88) In addition, the insulating member **6** includes the insulating wall portion **61** that provides insulation between the first wire connecting portion **44** of the center terminal **4** and the second wire connecting portion **52** of the outer terminal **5**, and the insulating wall portion **61** is provided in a manner extending from the insulating cylinder portion **60**. As a result, insulation can be provided between the first wire connecting portion **44** and the second wire connecting portion **52** with such a practical and simple structure using the insulating wall portion **61** that is provided in a manner

extending from the insulating cylinder portion **60**.

(89) Also, as shown in FIG. **4**, when the center terminal **4**, the outer terminal **5**, and the insulating member **6** are unitized, assembling work can be performed easily.

(90) The adapter housing **7** includes the to-be-positioned portions **79** (see FIG. **15**) arranged to engage the positioning grooves **104**, **105** of the opposing housing **101** so that the first intermediate contact **8** and the second intermediate contact **9** are aligned, respectively, with the corresponding opposing terminals **102**, **103**. As a result, when preliminarily fitting the adapter **1** to the opposing connector **100**, each of the intermediate contacts **8**, **9** is easily positioned with respect to the corresponding opposing terminals **102**, **103**.

(91) The present invention is not limited to the above-described preferred embodiments. For example, an insulating portion interposed between the center terminal portion **40** and the outer terminal portion **50** may be formed by a part of the connector housing **3**, although not shown.

(92) Two outer terminals **5**P, **50** may also be provided as in a modification shown in FIG. **18**. An outer terminal portion **50**P of the outer terminal **5**P and an outer terminal portion **50**Q of the outer terminal **50** are located apart in the axial direction (fitting direction X**1**) of the insulating cylinder portion **60** in a state enclosing the insulating cylinder portion **60**. In the adapter **1** side, two second intermediate contacts **9**P, **9**Q are provided correspondingly to the two respective outer terminals **5**P, **5**Q.

(93) In addition, the position of the contact portion **92**Pc of the fourth contact piece portion **92**P of the second intermediate contact **9**P with respect to the outer terminal **5**P and the position of the contact portion **92**Qc of the fourth contact piece portion **92**Q of the second intermediate contact **90** with respect to the outer terminal **5**Q are offset with respect to each other in the fitting direction X**1**. In this case, three-pole terminal connection is made possible regardless of the orientation of the connector with respect to the opposing connector. It is noted that three or more outer terminals and three or more second intermediate contacts may be provided.

(94) Although the present invention has been described in detail from the specific aspects, those skilled in the art who have understood the aforementioned content will easily recognize its modifications, variations, and equivalents. Therefore, the present invention should be within the scope of the claims and the scope of its equivalents.

REFERENCE SIGNS LIST

(95) **1**: Adapter **2**: Connector **3**: Connector housing **4**: Center terminal **5**; **5**P, **5**Q: Outer terminal **6**: Insulating member **7**: Adapter housing **8**: First intermediate contact **9**; **9**P, **9**Q: Second intermediate contact **30**: Main body portion **31**: Cylindrical portion **31***a*: Slit **31***b*: Inner peripheral surface **31***c*: Locking protrusion **32**: Cover **34**: Terminal portion housing portion **35**: Connecting space portion **40**: Center terminal portion **40***a*: Protruding portion **40***b*: Outer peripheral surface **43**: Positioning wall portion **44**: First wire connecting portion **50**; **50**P, **50**Q: Outer terminal portion **50***a*: Outer peripheral surface **51**: Positioning wall portion **52**: Second wire connecting portion **60**: Insulating cylinder portion (insulating portion) **61**: Insulating wall portion **66**: Engagement protrusion **73**: Fitting cylinder portion **74**: Engagement protrusion **75**: First intermediate contact retaining groove **76**: Second intermediate contact retaining groove **78**: Elastic piece portion **79**: To-be-positioned portion **80**: Fixed portion **81**: First contact piece portion **81***c*: Contact portion **82**: Second contact piece portion **82***c*: Contact portion **90**: Fixed portion **91**: Third contact piece portion **91***c*: Contact portion **92**; **92**P, **92**Q: Fourth contact piece portion **92***c*; **92**Pc, **92**Qc: Contact portion **100**: Opposing connector **101**: Opposing housing **102**, **103**: Opposing terminal **104**, **105**: Positioning groove (positioning portion) C**1**: Central axis CS: Connector structure D**1**, D**2**: Wire X**1**: Fitting direction

Claims

1. A connector structure comprising: an adapter to be fitted to an opposing connector that includes an opposing housing and a plurality of opposing terminals supported on the opposing housing and extending in parallel with each other; and a connector to be fitted to the adapter in a fitting direction, wherein the connector includes a connector housing, a center terminal supported on the connector housing and including a pin-shaped center terminal portion with a central axis extending in the fitting direction, and an outer terminal supported on the connector housing and including a cylindrical outer terminal portion that concentrically encloses the center terminal portion with an interposed insulating portion, the center terminal portion including a protruding portion that protrudes from the outer terminal portion in the fitting direction, the adapter includes an adapter housing to be fitted to the connector housing in the fitting direction and displaceable rotationally relative to the connector housing about the central axis and a first intermediate contact and a second intermediate contact supported on the adapter housing, the first intermediate contact includes a first contact piece portion to elastically contact the outer peripheral surface of the protruding portion of the center terminal portion and a second contact piece portion to elastically contact a corresponding one of the opposing terminals, and the second intermediate contact includes a third contact piece portion to elastically contact the outer peripheral surface of the outer terminal portion and a fourth contact piece portion to elastically contact a corresponding one of the opposing terminals.
2. The connector structure according to claim 1, wherein when the adapter housing and the connector housing are displaced relatively and rotationally with respect to each other, a contact portion of the first contact piece portion is displaced circumferentially in a sliding manner with respect to the outer peripheral surface of the center terminal portion and a contact portion of the third contact piece portion is displaced circumferentially in a sliding manner with respect to the outer peripheral surface of the outer terminal portion.
3. The connector structure according to claim 1, wherein the adapter housing includes a fitting cylinder portion that has an annular engagement protrusion extending entirely circumferentially on the outer peripheral surface of the adapter housing, the connector housing includes a cylindrical portion centered on the central axis, the cylindrical portion split circumferentially with a slit that extends axially to the tip end to be radially and elastically deformable and having at least one locking protrusion on the inner peripheral surface of the cylindrical portion, and when the fitting cylinder portion is inserted and fitted into the cylindrical portion, the locking protrusion overrides and engages the engagement protrusion so that the fitting cylinder portion is prevented from coming off.
4. The connector structure according to claim 1, wherein a plurality of the outer terminals are provided in a mutually insulated manner, a plurality of the second intermediate contacts are provided correspondingly to the plurality of outer terminals, and a contact portion of each of the plurality of second intermediate contacts and a corresponding one of the outer terminals are positionally offset with respect to each other in the fitting direction.
5. The connector structure according to claim 1, wherein the insulating portion includes an insulating member including an insulating cylinder portion into which the center terminal portion is inserted and fitted, and which is inserted and fitted into the outer terminal portion, the insulation cylinder portion being centered on the central axis.
6. The connector structure according to claim 5, wherein the center terminal includes a first wire connecting portion extending from the center terminal portion and to which a corresponding wire is connected, the outer terminal includes a second wire connecting portion extending from the outer terminal portion and to which a corresponding wire is connected, and the insulating member includes an insulating wall portion extending from the insulating cylinder portion and providing insulation between the first wire connecting portion and the second wire connecting portion.
7. The connector structure according to claim 5, wherein the center terminal, the outer terminal, and the insulating member are unitized.

8. The connector structure according to claim 1, wherein the adapter housing includes a to-be-positioned portion to engage a positioning portion of the opposing housing so that the first intermediate contact and the second intermediate contact are aligned, respectively, with corresponding ones of the opposing terminals.